Zijun Zhuang

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EDUCATION

University of Southern California — Los Angeles, CA
M.S. in Computer Science (Scientists and Engineers) — *Jan 2027 – Dec 2028 (Expected)*

University of California, Irvine — Irvine, CA
B.S. in Applied and Computational Mathematics — *Sep 2023 – Dec 2026*

- **Honors:** Dean's Honor List
- **Relevant Coursework:** Machine Learning, Probability, Linear Algebra, Numerical Analysis, Mathematical Modeling, ODE & PDE, Cryptology

RESEARCH EXPERIENCE

Dynamic Load Management for Data Centers — Irvine, CA
Research Assistant, CalPlug, Calit2; Advisor: Prof. G.P. Li — *Mar 2026 – Present*

- Built a Hyper-V testbed modeling a 50 kW, 18-rack edge facility, logging CPU utilization every 5 seconds alongside temperature, airflow, and power telemetry.
- Developed regression models predicting server energy draw from CPU utilization, establishing the group's first predictive baseline for comparison with fixed-setpoint cooling control; presented at the CalPlug/ITAC Workshop poster session.

Near-Isometry of Gaussian Random Matrices — Irvine, CA
Student Researcher, Directed Reading Program, Dept. of Mathematics — *Mar 2026 – Jun 2026*

- Wrote an 8-page paper on Gaussian random matrix near-isometries, deriving concentration bounds for norm and inner-product preservation.
- Validated the distortion bounds in Python across four random-matrix ensembles and constructed a counterexample showing isotropy is necessary; presented a poster.

Self-Correction in LLM Grading Agents — Irvine, CA
Independent Project — *May 2025 – Jun 2025*

- Built a reference-free grading agent (Qwen2.5-1.5B, vLLM) that judges GSM8K solutions without an answer key and iteratively revises its verdicts through multi-round self-critique.
- Found no meaningful accuracy improvement from intrinsic self-correction: across 300 solutions evaluated over three critique rounds, grader decisions changed 77 times, with final accuracy of 47.3% versus 47.0% for single-pass grading (44% of solutions were correct).
- Identified a leniency-drift failure mode: critique rounds biased the grader toward accepting answers, reducing the wrong-answer detection rate from 21% to 17%, consistent with prior work on the limits of intrinsic self-correction.

Automated Grading for Handwritten Mathematics — Irvine, CA
Research Assistant, Dept. of Mathematics; Advisor: Prof. Yifeng Yu — *Jan 2025 – Jun 2025*

- Contributed to an automated scoring and feedback system for handwritten calculus exams; preprocessed scans with OpenCV and reimplemented U-Net, ResNet, and Transformer in PyTorch for document image processing.
- Used GPT-4o via the OpenAI API with prompt engineering to generate rubric-based scores and written feedback; annotated scanned exams to build the training dataset and presented weekly literature reviews.

INTERNSHIP EXPERIENCE

Tencent Technology (Shanghai) Co., Ltd. — Shanghai, China
Operations Intern — *Jul 2025 – Aug 2025*

- Tracked 300+ monthly IDC maintenance work orders across 5 contractors; built a Python pipeline consolidating records from dozens of suppliers into a priority-ranked weekly report for management.

Shanghai Youxun Industry Co., Ltd. — Shanghai, China
IT & Business Intern — *Jul 2024 – Sep 2024*

- Supported network deployment (fiber, MPLS VPN, SD-WAN) for 20+ corporate clients and served as an English–Mandarin liaison during five cross-border client presentations.

SKILLS

- **Programming:** Python (NumPy, Pandas), C++, Java, R, MATLAB
- **Machine Learning:** PyTorch, TensorFlow, Hugging Face, vLLM, OpenAI API, OpenCV
- **Tools & Systems:** Git, Linux, Microsoft Hyper-V
- **Languages:** English, Mandarin